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Seasonal controls on groundwater irrigation quality in a semi-arid granitic watershed of Telangana, India: an integrated *IWQI*-GIS-multivariate approach

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Groundwater is a vital resource for irrigation in arid and semi-arid regions. This study evaluates groundwater suitability for irrigation in the granitic watershed of Dilawarpur Mandal, Nirmal District, Telangana, India, using 61 groundwater samples collected during pre- and post-monsoon seasons. Major physicochemical parameters (pH, EC, TDS, Ca^{2+}, Mg^{2+}, Na^{+}, K^{+}, HCO_3^{-}, Cl^{-}, SO_4^{2-}, and F^{-}) were analysed. Irrigation suitability was assessed using the Irrigation Water Quality Index (*IWQI*) in conjunction with EC, *%Na⁺*, *SAR*, *SSP*, *KR*, *MHR*, *PI*, and *RSC*. Multivariate statistical techniques, including Pearson correlation, Principal Component Analysis (PCA), Hierarchical Cluster Analysis (HCA), and Multiple Linear Regression (MLR), were applied to identify dominant hydrochemical controls and evaluate predictive relationships. Although elevated EC indicates moderate to high salinity at several locations, most groundwater samples are suitable for irrigation. *IWQI* results show that 64% of pre-monsoon and 66% of post-monsoon samples fall within excellent to good categories, with post-monsoon improvement attributed to monsoonal dilution of dissolved salts and Na^{+}. Spatial *IWQI* analysis indicates an expansion of excellent-quality zones from 85% in pre-monsoon to 96% in post-monsoon season. Correlation analysis, PCA, and HCA consistently identify Na^{+}-related indices and salinity as the primary controls on irrigation suitability, while carbonate balance and soil permeability act as secondary modifiers. PCA explains 92.21% and 87.57% of the total variance during pre- and post-monsoon seasons, respectively. MLR reveals stronger and more predictable hydrochemical control during the pre-monsoon season (R^2 = 0.71) compared to the post-monsoon season (R^2 = 0.33), where dilution and hydrochemical mixing reduce model predictability. The study highlights Na^{+}-related processes as the principal constraint on irrigation water quality and emphasizes the need for season-specific groundwater management strategies in semi-arid hard-rock terrains.

Keywords Groundwater quality, *IWQI*, Seasonal variation, Multivariate analysis, GIS, Granite terrain

Groundwater is often the only reliable source of freshwater in arid and semi-arid regions, where surface water availability is limited, intermittent, or highly seasonal[1,2]. In these environments, groundwater is critical for supporting domestic needs, livestock, and irrigated agriculture, thereby underpinning food security and rural livelihoods[3,4]. However, rapid increases in groundwater abstraction, increased fertilizer use, and inefficient irrigation methods have gradually changed aquifer hydrochemistry[5,6]. These pressures typically accelerate salinization and sodium (Na^{+}) enrichment, leading to soil clay dispersion, reduced permeability, impaired infiltration, and long-term declines in agricultural productivity[7–9]. Consequently, a systematic and seasonally consistent assessment of groundwater suitability for irrigation has become vital for sustainable agricultural planning in semi-arid regions[10,11].

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Irrigation water suitability is traditionally evaluated using a suite of hydrochemical indices, including Electrical Conductivity (EC), Sodium Adsorption Ratio (*SAR*), Sodium Percentage (*%Na*$^+$), Kelly's Ratio (*KR*), Soluble Sodium Percentage (*SSP*), Magnesium Hazard Ratio (*MHR*), Permeability Index (*PI*), and Residual Sodium Carbonate (*RSC*). Each index reflects specific soil–water interaction mechanisms, such as osmotic stress, sodicity-induced dispersion, or carbonate-related permeability reduction[12,13]. However, independent interpretation of these indices often yields fragmented or contradictory assessments, particularly under seasonally variable hydrogeochemical conditions. Recent studies have therefore emphasized the need for integrated evaluation frameworks, such as the Irrigation Water Quality Index (IWQI). The IWQI synthesizes multiple parameters into a single management-oriented metric and improves the spatial and temporal discrimination of irrigation suitability[14,15].

Hard-rock aquifers in southern India, particularly granitic terrains, are especially vulnerable to irrigation-related salinity and sodium hazards due to low primary porosity, restricted fracture connectivity, limited buffering capacity, and long groundwater residence times[16,17]. In these aquifers, monsoonal recharge plays a dominant role in regulating groundwater chemistry through dilution, ion exchange, and mixing processes, often resulting in short-term improvements in irrigation water quality[18]. However, emerging evidence suggests that dilution alone may not fundamentally alter sodium-dominated hydrochemical facies or carbonate-related hazards. Consequently, soil degradation risks may persist even during post-monsoon periods[19].

Despite extensive groundwater quality research in Telangana and adjoining semi-arid regions, several critical knowledge gaps remain insufficiently addressed. Most existing studies emphasize drinking water suitability or rely on single-season datasets, thereby failing to capture seasonal hydrochemical dynamics relevant to irrigation management[20–22]. As a result: (i) seasonal variability in salinity, sodicity, and carbonate-related irrigation hazards remains inadequately quantified; (ii) the hierarchical interdependence among irrigation indices under contrasting hydrological conditions is poorly constrained; (iii) multivariate statistical groupings are rarely interpreted in terms of underlying hydrogeochemical processes; and (iv) predictive frameworks capable of translating hydrochemical variability into irrigation-relevant decision support are limited.

Addressing these gaps is crucial because irrigation management decisions based on incomplete or seasonally biased assessments may underestimate sodium-induced soil degradation and compromise long-term agricultural sustainability.

Previous studies from the study region have shown limitations in single-season assessments of groundwater quality for drinking and irrigation[23–25] and in the temporal dynamics of groundwater quality and associated health risks[26]To address these limitations, the present study conducts a comprehensive spatiotemporal assessment of irrigation water quality in a semi-arid granitic watershed in Dilawarpur Mandal, Nirmal District, Telangana. This study integrates classical hydrochemical analysis with multivariate and predictive statistical methods, including Pearson's correlation (r), PCA, HCA, and MLR. PCA and HCA identify dominant hydrogeochemical controls and parameter associations, while MLR quantifies the influence of key variables on IWQI and evaluates seasonal variations in predictive stability and explanatory power[27–29]. This descriptive–predictive framework enables robust interpretation of controlling processes and their seasonal modulation.

Furthermore, the study examines whether monsoonal recharge fundamentally restructures sodium-dominated hydrochemical regimes in hard-rock aquifers or merely induces transient dilution effects. By explicitly linking recharge-driven chemical transformations to multivariate structure and predictive modeling, this work advances a process-based understanding of the evolution of irrigation water quality in semi-arid granitic terrains.

Based on the available literature, such a seasonally resolved, predictive *IWQI*–multivariate framework has not been previously applied to semi-arid granitic watersheds in Telangana. Accordingly, the specific objectives of this study are to:

a) Evaluate pre- and post-monsoon variations in major irrigation water quality parameters (EC, *%Na*$^+$, *SAR, KR, MHR, SSP, PI, RSC*, and *IWQI*);
b) Identify dominant hydrogeochemical processes and parameter interrelationships using correlation coefficient, PCA, HCA, and MLR;
c) Quantify the influence of monsoonal recharge on Na^+ dynamics, carbonate hazards, soil permeability, and seasonal predictive stability of irrigation water quality; and
d) Delineate irrigation management zones and propose site-specific mitigation strategies based on multivariate clustering and spatial *IWQI* patterns.

By explicitly addressing key methodological and conceptual gaps in seasonal irrigation water quality assessment, this study provides a decision-oriented framework for sustainable groundwater management in hard-rock aquifers. The findings provide new insights into the persistence of sodium–salinity dominance under monsoon-driven hydrological variability. These results support the development of sustainable irrigation strategies to protect soil health, maintain crop productivity, and preserve long-term groundwater resources, thereby contributing to Sustainable Development Goals (SDG 2 and SDG 6).

Study area

Location and climate

The study was conducted in the Dilawarpur Mandal watershed of Nirmal District, northern Telangana, India, covering an area of approximately 156 km^2 (Fig. 1). The watershed lies between latitudes 19°00′–19°09′ N and longitudes 78°12′–78°18′ E and is represented on Survey of India topographic sheets 56 I/4, 56 I/8, and 56 J/5 at a scale of 1: 50,000.

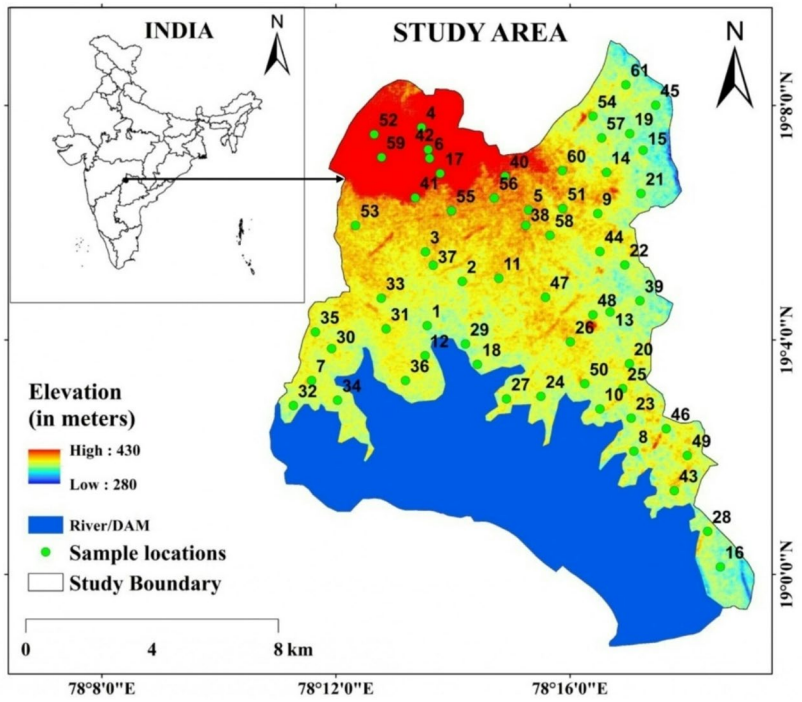


Fig. 1. Location map of the study area showing groundwater sampling locations in Dilawarpur Mandal, Telangana, India. The base map was prepared using Survey of India topographical sheets. The India national boundary shape file was obtained from the NRSC Bhuvan, and the DEM data used for elevation mapping were sourced from the USGS Earth Explorer. All spatial data were processed and maps were prepared using ArcGIS 10.8 (Esri Inc.).

The region experiences a semi-arid climate with pronounced seasonal variability. Summer temperatures frequently reach ~ 45 °C, while winter temperatures remain relatively mild, with minima around 22 °C. Mean annual rainfall during 2011–2021 is approximately 1,022 mm, most of which occurs during the southwest monsoon[26,30]. High summer temperatures combined with uneven rainfall distribution result in elevated evapotranspiration rates and recurrent seasonal water stress.

Geology and hydrogeology

Geologically, the study area forms part of the Archean crystalline terrain of Nirmal District, dominated by lithounits of the unclassified Dharwar Supergroup. The bedrock mainly consists of the Peninsular Gneissic Complex, including pink hornblende–biotite granite, pink granite, and grey hornblende–biotite gneiss, which are widely exposed across the watershed (Fig. 2a). These basement rocks are intruded by Paleoproterozoic dolerite dykes trending predominantly north–south and northeast–southwest.

Limited exposures of younger sedimentary formations belonging to the Pahal Super group (Mesoproterozoic) and the Penganga Group (Neoproterozoic) occur locally along lithological contacts. These units comprise quartzite, shale, and limestone and unconformably overlie the crystalline basement. The predominance of granitic and doleritic lithologies, which commonly host fluoride-bearing minerals such as fluorite, fluorapatite, and biotite, exerts a strong control on regional groundwater geochemistry.

Hydrogeologically, groundwater occurrence is primarily controlled by secondary porosity generated by weathering and rock fracturing. Productive aquifers are largely confined to the weathered mantle and the underlying fractured bedrock, with groundwater abstraction commonly extending to depths greater than 65 m. Spatial variability in the thickness of the weathered zone, fracture intensity, and structural interconnectivity governs groundwater storage and flow dynamics, as well as residence time, thereby enhancing water–rock interactions and exerting a strong influence on the evolution of major ion chemistry.

The predominant crops include paddy, maize, and pulses. Irrigation is mainly dependent on groundwater through borewells and shallow wells, supplemented by seasonal canal water where available. These practices strongly influence local groundwater demand and quality.

Land-use and land-cover (LULC)

Land-use and land-cover (LULC) characteristics were derived using Survey of India topographic sheets (56 I/4, 56 I/8, and 56J/5), IRS-P6 (2010) satellite imagery, and recent false-color composite (FCC) images from 2020 (Fig. 2b). Data processing and interpretation were carried out with support from the National Remote Sensing Agency (NRSA), Hyderabad, India.

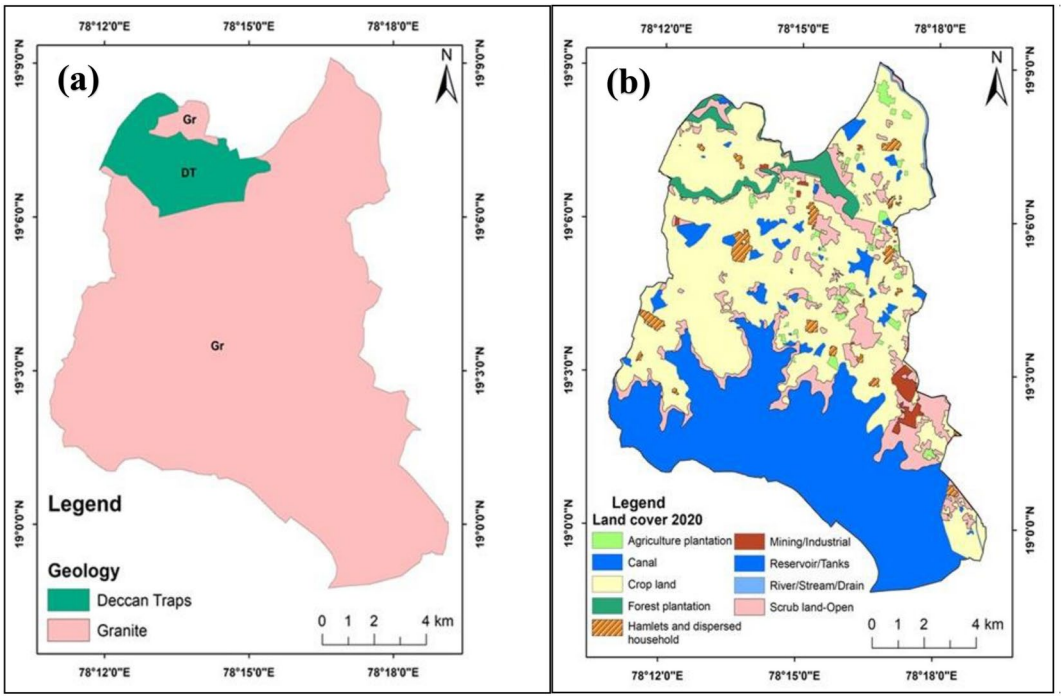


Fig. 2. (**a**) Geological map, and (**b**) Land-use/land-cover distribution.

The LULC classification indicates that agricultural land (48.27%) is the dominant land use, followed by water bodies (35.99%), shrub land (10.50%), forest cover (2.66%), built-up areas (1.60%), and fallow land (0.98%). These land-use patterns play a critical role in controlling spatial variability in groundwater recharge, abstraction intensity, and irrigation-related water quality across the watershed.

Materials and methodology

Sampling, analytical procedures and ionic balance

A total of 61 groundwater samples were collected from borewells across Dilawarpur Mandal during pre-monsoon and post-monsoon seasons 2020 to capture seasonal hydrochemical variability. Sampling was conducted using pre-cleaned 1 L high-density polyethylene (HDPE) bottles following APHA standard protocols[31]. Samples were immediately sealed, labelled, and transported to the Centre for Materials for Electronics Technology (C-MET), Hyderabad, India, for laboratory analysis.

In situ measurements of pH and electrical conductivity (EC) were carried out using calibrated portable meters (HACH SensION and Systronics Conductivity Meter, Model 304). Total Dissolved Solids (TDS) were estimated from EC using standard conversion factors.

Major cations (Ca^{2+}, Mg^{2+}, Na^{+}, and K^{+}) and anions (HCO_3^{-}, Cl^{-}, SO_4^{2-}, NO_3^{-}, and F^{-}) were analyzed following APHA and BIS guidelines. Na^{+} and K^{+} were determined using flame photometry (Systronics Model 128), while Ca^{2+} and Mg^{2+} were measured by EDTA titration. HCO_3^{-} was analyzed using standard titrimetric methods.

Analytical reliability was assessed using the charge balance error (*CBE*) between total cations and anions (Eq. 1):

$$BE = \frac{\sum Cations - \sum Anions}{\sum Cations + \sum Anions} \times 100 \quad (1)$$

Only samples with CBE within ± 5% were considered acceptable, ensuring analytical consistency and data reliability. In the present study area, it was observed within this limit (2.3 to 4.5%).

Irrigation water quality indices

Groundwater suitability for irrigation was evaluated using hydrochemical indices that reflect salinity, sodicity, alkalinity, and soil permeability hazards. These include EC, *%Na*$^{+}$, *SAR, KR, SSP, RSC, PI, MHR*, and the Irrigation Water Quality Index (*IWQI*). Each index captures a specific soil–water interaction, and using them together reduces uncertainty from relying on a single index.

Sodium hazard indices (EC, ***%Na***$^{+}$*,* ***SAR****,* ***KR****, and* ***SSP****)*

Electrical conductivity (EC) was used as a primary indicator of salinity hazard and classified according to standard irrigation water quality criteria. Sodium-related indices (%Na^{+}, SAR, KR, and SSP) were calculated

using ionic concentrations expressed in mill equivalents per liter (meq/L) to assess sodicity-related risks affecting soil structure and permeability, following the methods proposed by Wilcox[32], Richards[33], and Kelly[34].

The percentage sodium (*%Na*$^+$) measures the proportion of Na^+ among major cations (Eq. 2). High Na^+ can reduce soil permeability and hinder plant growth.

$$\%Na = \frac{Na^+}{Ca^{2+} + Mg^{2+} + Na^+ + K^+} \times 100 \tag{2}$$

According to Wilcox[27], %Na^+ < 60% is suitable; > 60% indicates sodicity risk.

SAR indicates the dominance of Na^+ over Ca^{2+} and Mg^{2+} (Eq. 3), affecting soil structure and infiltration:

$$SAR = \frac{Na^+}{\sqrt{\frac{1}{2}(Ca^{2+} + Mg^{2+})}} \tag{3}$$

SAR classification categorizes water as Excellent (*SAR* < 10), Good (10–18), Doubtful (18–26), and Unsuitable (> 26).

Kelly's ratio (*KR*) compares Na^+ to Ca^{2+} + Mg^{2+} (Eq. 4):

$$KR = \frac{Na^+}{Ca^{2+} + Mg^{2+}} \tag{4}$$

KR < 1 indicates suitability; *KR* > 1 suggests Na^+ dominance and reduced permeability.

Soluble Sodium Percentage (*SSP*) assesses Na^+ accumulation risk (Eq. 5):

$$SSP = \frac{Na^+}{Ca^{2+} + Mg^{2+} + Na^+} \tag{5}$$

SSP < 25% is suitable; > 25% indicates potential long-term sodicity hazards.

Carbonate hazard indices (RSC, PI, and MHR)

RSC evaluates the impact of CO_3^{2-} and HCO_3^+ on Ca^{2+} and Mg^{2+} precipitation (Eq. 6), which can increase soil alkalinity:

$$RSC = \left[HCO_3^- + CO_3^-\right] - \left[Ca^{2+} + Mg^{2+}\right] \tag{6}$$

Groundwater is classified as Suitable (*RSC* < 1.25), Doubtful (1.25–2.50), Unsuitable (> 2.50)[28].

Permeability Index (*PI*) reflects how Na^+, Ca^{2+}, Mg^{2+}, and HCO_3^- affect soil permeability (Eq. 7):

$$PI = \frac{N(a^{2+} + \sqrt{HCO_3^-})}{(Ca^{2+} + Mg^{2+} + Na^+} \times 100 \tag{7}$$

PI classes: Class I (> 75%) suitable; Class II (25–75%) moderate; Class III (< 25%) unsuitable.

Magnesium Hazard Ratio (*MHR*) indicates Mg^{2+} dominance (Eq. 8):

$$MHR = \frac{Mg^{2+}}{Ca^{2+} + Mg^{2+}} \times 100 \tag{8}$$

MHR < 50 is suitable; *MHR* > 50 indicates reduced soil quality and productivity.

Irrigation water quality index (*IWQI*)

IWQI integrates EC, *SAR*, Na^+, Cl^-, and HCO_3^- to provide an overall measure of irrigation suitability[35].

Na^+, Cl^-, and HCO_3^- were converted from mg/L to meq/L, and dimensionless sub-indices (*qi*) were calculated (Eq. 9):

$$q_i = q_{imax} - [(x_{ij} - x_{inf})xq_{iamp}]/q_{amp} \tag{9}$$

where *qi* is the sub-index of the *i*th water quality parameter, q_{imax} is its maximum sub-index value, *xij* is the measured concentration of the *i*th parameter in the *j*th groundwater sample, x_{inf} is the ideal (lower threshold) concentration, Xq_{iamp} represents the sub-index range, and x_{amp} denotes the concentration range between the upper and lower threshold limits of the *i*th parameter.

The relative weight (*Wi*) assigned to each parameter reflects its relative influence on irrigation suitability (Eq. 10).

$$W_i = \frac{\sum_{j=1}^{k} F_j A_{ij}}{\sum_{j=1}^{k} \sum_{i=1}^{n} F_j A_{ij}} \tag{10}$$

Parameters	W_i
EC	0.211
Na^+	0.204
HCO_3^-	0.202
Cl^-	0.194
SAR	0.189
ΣWi	1.000

Table 1. Relative weights (W_i) assigned to *IWQI* parameters.

Parameters	Range of calculations of *IWQI*			
q_i	85–100	60–85	35–60	0–35
EC	200 < EC < 750	750 < EC < 1500	1500 < EC < 3000	EC < 200 or EC > 3000
SAR	2 < *SAR* < 3	3 < *SAR* < 6	6 < *SAR* < 12	*SAR* < 2 or *SAR* > 12
Na^+	2 < Na^+ < 3	3 < Na^+ < 6	6 < Na^+ < 12	Na^+ < 2 or Na^+ > 2
Cl	1 < Cl^- < 4	4 < Cl^- < 7	7 < Cl^- < 10	Cl^- < 1 or Cl^- > 12
HCO_3^-	1 < HCO_3^- < 1.5	1.5 < HCO_3^- < 4.5	4.5 < HCO_3^- < 8.5	HCO_3^- > 1 or HCO_3^- > 8.5

Table 2. Classification ranges and sub-index scores used for *IWQI* computation[30].

where *Wi* is the normalized weight assigned to the *i*th water quality parameter (Table 1), *Fj* is the factor loading of the *j*th principal component, *Aij* represents the contribution of the *i*th parameter to the *j*th principal component, *i* (1–*n*) denotes the water quality parameter index, *j* (1–*k*) denotes the principal component number, *n* is the total number of parameters used in the *IWQI* computation, and *k* is the total number of retained principal components.

The final *IWQI* was calculated using weighted aggregation (Eq. 11):

$$IWQI = \sum_{j=1}^{k} q_i X w_i \tag{11}$$

where *IWQI* is the irrigation water quality index, *qi* is the sub-index value of the *i*th parameter, W_j is the weight assigned to the *i*th parameter, and *n* is the number of parameters used in *IWQI* calculation.

The classification of *IWQI* computation is shown in Table 2.

Statistical and multivariate analysis

Pearson correlation coefficient

Pearson correlation coefficients (*r*) were calculated among all irrigation indices (EC, *%Na*$^+$, *SAR*, *KR*, *MH*, *PI*, *SSP*, *RSC*, and *IWQI*) for pre-monsoon and post-monsoon datasets. Positive "*r*" values indicate a direct relationship, while negative values indicate an inverse relationship[36].

All variables were standardized prior to analysis to ensure uniform scaling. Statistical significance was assessed at the 95% and 99% confidence levels ($p < 0.05$ and $p < 0.01$). Correlation analysis identifies strongly interdependent indices, especially those related to salinity and sodicity, helping to clarify their combined effect on overall irrigation water quality.

Principal component analysis (PCA)

PCA was applied to reduce dataset dimensionality and identify dominant hydrochemical processes controlling groundwater suitability[37]. All indices were standardized (mean = 0, SD = 1) to remove scale effects. Eigenvalues and eigenvectors were extracted from the covariance matrix, and PCs with eigenvalues > 1 were retained following the Kaiser criterion. Loadings of variables on each PC were used to quantify their contribution to total variance and distinguish primary controlling factors from secondary influences.

Hierarchical cluster analysis (HCA)

HCA was used to group irrigation water quality indices based on similarity in behavior and influence on groundwater suitability. Ward's linkage method with squared Euclidean distance was used as the dissimilarity measure. Standardization of all variables minimized scale bias. HCA provides insights into index groupings and their hydrochemical significance, facilitating interpretation of groundwater quality patterns.

Multiple linear regression (MLR)

To evaluate the relative contribution and seasonal sensitivity of key hydrochemical parameters influencing irrigation water quality, Multiple Linear Regression (MLR) analysis was performed separately for the pre-monsoon and post-monsoon datasets. In this study, MLR was applied as a diagnostic and sensitivity-based

tool, rather than as an independent predictive model, to examine how the influence of major irrigation-relevant parameters varies under contrasting seasonal hydrogeochemical conditions.

The *IWQI* was treated as the dependent variable, while *EC*, Na^+, HCO_3^-, Cl^-, and *SAR* were selected as independent variables due to their direct relevance to salinity, sodicity, and alkalinity hazards affecting soil permeability and crop productivity.

The regression model is expressed as (Eq. 12):

$$IWQI = \beta 0 + \beta 1\,(EC) + \beta 2(Na^+) + \beta 3\left(HCO_3^-\right) + \beta 4\left(Cl^-\right) + \beta 5\,(SAR) + \varepsilon \tag{12}$$

where IWQI is the Irrigation Water Quality Index (dimensionless), β_0 is the intercept (constant term) of the regression/index model, β_1–β_5 are the model coefficients (weights) assigned to each parameter, EC is the electrical conductivity of groundwater (µS/cm), Na^+ os the sodium concentration in water (mg/L, as used in your dataset), HCO_3^- is the bicarbonate concentration (mg/L), Cl^- is the chloride concentration (mg/L), SAR is the sodium adsorption ratio (dimensionless), and ε is the random error/residual term of the model.

MLR models were developed for pre-monsoon and post-monsoon conditions to assess seasonal variations in regression coefficients, parameter significance, and explanatory strength, thereby highlighting changes in the hydrochemical controls on irrigation water quality associated with monsoonal recharge.

Model performance was evaluated using standard statistical indicators, including the coefficient of determination (R^2), adjusted R^2, root mean square error (*RMSE*), and mean absolute error (*MAE*) (Eqs. 13–15). These metrics were used to assess goodness-of-fit and parameter sensitivity, rather than to infer independent predictive capability:

$$R^2 = 1 - \frac{\sum (O_{i-}P_i)^2}{\sum (O_i - \overline{O}_i)^2} \tag{13}$$

$$RMSE = \sqrt{\frac{\sum (O_i - P_i)^2}{n}} \tag{14}$$

$$MAE = \frac{\sum IO_i - P_iI}{n} \tag{15}$$

where *O* denotes the observed IWQI value, *P* denotes the estimated IWQI value, and *n* represents the total number of samples.

The R^2 reflects the proportion of *IWQI* variance explained by the contributing parameters, while the adjusted R^2 accounts for model complexity. The *F*-statistic and corresponding *p*-values were used to evaluate overall model significance. *RMSE* and *MAE* provide complementary measures of residual dispersion and model stability, supporting interpretation of parameter influence rather than standalone prediction[38].

All regression and correlation analyses were performed using STATISTICA (version 6), and statistical significance was evaluated at the 99% confidence level ($p < 0.01$).

Geospatial mapping

Spatial distribution maps were prepared using the Inverse Distance Weighting (IDW) interpolation approach in ArcGIS 10.8. This method was selected based on the principle that spatially proximate sampling locations tend to exhibit comparable groundwater quality characteristics, allowing for reliable estimation of unsampled areas. A fixed search radius was applied, and the power parameter was adjusted to achieve optimal interpolation performance by reducing prediction errors. The generated maps clearly depict the spatial variability and delineate zones of varying groundwater quality across the study region.

Results and discussion

Hydrochemical characteristics

Groundwater pH remains slightly alkaline in both seasons (7.02–8.35 in pre-monsoon; 7.40–8.30 in post-monsoon), reflecting carbonate buffering processes (Table 3). Low CV values (< 5%) indicate geochemical stability and near-equilibrium conditions with carbonate minerals[39].

Electrical conductivity (EC) ranges from 414–2328 µS/cm (mean: 934 µS/cm) during pre-monsoon and 375–1594 µS/cm (mean: 751 µS/cm) during post-monsoon season (Table 3). Higher pre-monsoon EC reflects evapoconcentration, prolonged groundwater residence time, and intensive abstraction, whereas lower post-monsoon values indicate dilution due to monsoonal recharge[40]. Based on irrigation suitability criteria (Table 4), 45% (pre-monsoon) and 47% (post-monsoon) of samples fall within the good salinity class (250–750 µS/cm), while 50% and 53%, respectively, belong to the high salinity class (750–2250 µS/cm). Only 5% of pre-monsoon samples exceed 2250 µS/cm, indicating localized salinity accumulation prior to recharge.

Total dissolved solids (TDS) exhibit a pattern similar to EC, ranging from 265–1490 mg/L (mean: 598 mg/L) in pre-monsoon and 240–1020 mg/L (mean: 481 mg/L) in post-monsoon season (Table 3). Higher pre-monsoon variability (CV: 38.23%) compared to post-monsoon (30.55%) indicates spatial heterogeneity influenced by evapotranspiration and lithological control. This spatial variability becomes relatively homogenized after monsoonal recharge[41].

Among major cations, Ca^{2+} and Na^+ are dominant. Ca^{2+} concentrations range from 40–144 mg/L (mean: 69 mg/L) in pre-monsoon and 26–118 mg/L (mean: 59 mg/L) in post-monsoon season (Table 3), primarily derived from silicate weathering and limited carbonate dissolution[38]. Magnesium occurs in moderate

Chemical variables	Pre-monsoon					Post monsoon				
	Min	Max	Mean	SD	CV	Min	Max	Mean	SD	CV
pH (unitless)	7.02	8.35	7.64	0.30	4.32	7.40	8.30	7.96	0.24	3.07
EC (µS/cm)	414	2328	934	394.18	37.75	375	1594	751	248.02	30.49
TDS (mg/l)	265	1490	598	254.85	38.23	240	1020	481	158.23	30.55
Ca^{2+} (mg/l)	40	144	69	20.82	30.00	26	118	59	17.44	33.86
Mg^{2+} (mg/l)	18	118	37	19.17	52.32	14	64	30	9.82	3.07
Na^{+} (mg/l)	22	225	81	54.13	67.04	32	166	58	25.51	43.85
K^{+} (mg/l)	1	12	4	2.827	74.34	1	4	2	0.84	40.46
HCO_3^{-} (mg/l)	120	390	236	63.21	26.74	86	380	188	54.22	28.85
Cl^{-} (mg/l)	46	410	132	88.50	66.81	30	276	98	17.44	33.86
SO_4^{2-} (mg/l)	8	96	25	17.06	68.32	12	61	26	11.38	43.84
NO_3^{-} (mg/l)	3	135	39	31.40	81.10	6	132	39	24.71	64.08
F^{-} (mg/l)	0.24	3.28	1.01	0.61	60.26	0.08	1.52	0.65	0.41	64.07

Table 3. Statistical summary of major physicochemical parameters of groundwater during pre- and post-monsoon seasons. Min, Max, SD, and CV denote minimum, maximum, standard deviation, and coefficient of variance (%), respectively

ISI	Ranges	Classification of water	Pre-monsoon		Post-monsoon	
			No. of samples	% of samples	No. of samples	% of samples
Salinity hazard						
EC (µS/cm)	< 250	Excellent	-	-	-	-
	250 – 750	Good	27	45	29	47
	750 – 2250	High	31	50	32	53
	> 2250	Very high	3	5	-	-
Sodium hazard indices						
$\%Na^{+}$	< 60	Suitable	61	100	61	100
	> 60	Unsuitable	-	-	-	-
SAR	1 – 10	Excellent	61	100	61	100
	10 – 18	Good/	-	-	-	-
	18 – 26	Doubtful	-	-	-	-
	> 26	Unsuitable	-	-	-	-
KR	< 1	Suitable	58	95	61	100
	> 1	Unsuitable	3	5	-	-
	< 40	Unsuitable	-	-	-	-
SSP (%)	< 25	Suitable	47	77	9	15
	> 25	Unsuitable	14	23	52	85
Carbonate hazard indices						
RSC (meq/L)	< 1.25	Good	61	100	61	100
	1.25 – 2.50	Doubtful	-	-	-	-
	> 2.50	Unsuitable	-	-	-	-
PI (%)	> 75	Good	2	3	-	-
	25 – 75	Doubtable	59	97	61	100
	< 25	Unsuitable	-	-	-	-
MHR (%)	< 50	Suitable	52	85	58	95
	> 50	Not suitable	9	15	3	5
Irrigation water quality index						
IWQI (%)	> 85	Excellent	14	23	24	39
	60 – 85	Good	38	62	35	57
	35 – 0	Poor	3	5	2	4
	< 35	Very poor	6	10	-	-

Table 4. Classification of groundwater samples based on irrigation suitability indices (ISI) during pre- and post-monsoon seasons.

concentrations (18–118 mg/L in pre-monsoon and 14–64 mg/L in post-monsoon). The higher pre-monsoon variability (CV: 52.32%) indicates heterogeneous contributions from ferromagnesian minerals and ion exchange processes.

Sodium concentrations show wide spatial variation (22–225 mg/L in pre-monsoon; 32–166 mg/L in post-monsoon) and high CV values (67.04% and 43.85%, respectively), reflecting alkali feldspar weathering, cation exchange, and anthropogenic inputs such as irrigation return flow[42] (Table 3). Potassium remains low (< 12 mg/L) in both seasons due to fixation by clay minerals, though elevated CV values (74.34% in pre-monsoon and 40.46% in post-monsoon) suggest localized enrichment linked to agricultural activities.

Among anions, HCO_3^- predominates, ranging from 120–390 mg/L (mean: 236 mg/L) in pre-monsoon and 86–380 mg/L (mean: 188 mg/L) in post-monsoon season (Table 3). This dominance indicates CO_2-mediated silicate weathering as the principal geochemical process in the granitic aquifer. Chloride concentrations vary considerably (46–410 mg/L in pre-monsoon; 30–276 mg/L in post-monsoon), reflecting evaporative concentration, irrigation return flow, and localized domestic inputs[43].

Sulphate concentrations are generally low to moderate (8–96 mg/L in pre-monsoon; 12–61 mg/L in post-monsoon) and are attributed to minor sulphate mineral dissolution and agricultural sources[44]. Nitrate exhibits very high spatial variability (CV > 60% in both seasons), with concentrations reaching up to 135 mg/L in the pre-monsoon and 132 mg/L in the post-monsoon season. This indicates anthropogenic contamination from fertilizer application, sanitation practices, and shallow recharge pathways[45].

Fluoride concentrations range from 0.24–3.28 mg/L (mean: 1.01 mg/L) in pre-monsoon and 0.08–1.52 mg/L (mean: 0.65 mg/L) in post-monsoon season (Table 3). Elevated pre-monsoon levels reflect geogenic release from fluorite- and biotite-bearing minerals under alkaline conditions, enhanced by evapoconcentration.

Seasonal variation of irrigation water quality indices

Salinity hazard

Salinity hazard, evaluated using electrical conductivity (EC), is a critical factor governing irrigation water suitability because elevated salt concentrations reduce soil osmotic potential and restrict crop water uptake. The predominance of moderate to high salinity classes during pre-monsoon period reflects the cumulative effects of evapoconcentration, prolonged groundwater residence time, and intensive abstraction under semi-arid conditions[46]. The occurrence of very high salinity waters during this season indicates localized zones of hydraulic stagnation and limited natural flushing, which enhance solute accumulation in hard-rock aquifers.

Post-monsoon EC values generally shift toward lower salinity classes, demonstrating the diluting influence of monsoonal recharge[47]. However, the continued dominance of medium (C2) and high (C3) salinity classes in both seasons, based on the USSL classification[48] (Fig. 3), indicates that recharge results in partial dilution rather than complete chemical renewal of groundwater. This persistence highlights the resilience of the aquifer's geochemical framework, wherein seasonal recharge moderates' salinity levels without fundamentally altering long-term salinity controls.

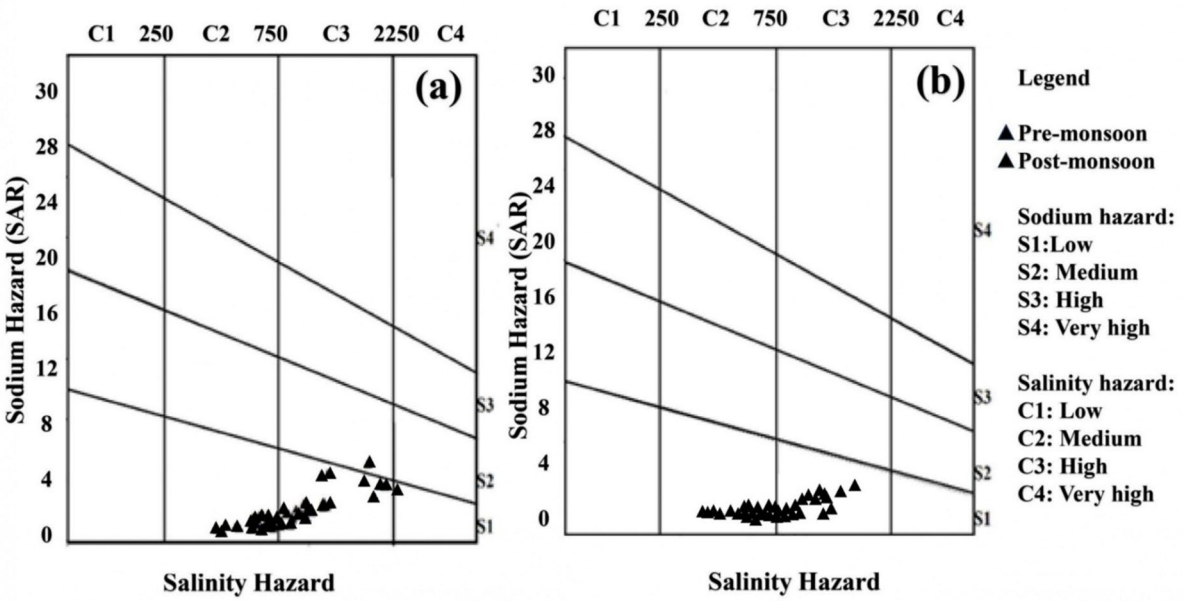


Fig. 3. USSL diagram showing irrigation water suitability during (**a**) pre-monsoon and (**b**) post-monsoon seasons.

Water quality indices	Pre-monsoon					Post monsoon				
	Min	Max	Mean	SD	CV	Min	Max	Mean	SD	CV
Salinity hazard										
EC (µS/cm)	414.00	2328	933.62	430.42	46.10	375	1594	751.21	258.06	34.35
Sodium hazard indices										
$\%Na^+$	16.17	57.98	32.48	09.18	28.01	21.10	43.82	31.46	05.72	18.18
SAR	0.61	5.08	1.88	1.00	53.45	0.86	3.16	1.52	0.47	31.19
KR	0.19	1.46	0.52	0.25	48.33	0.27	0.79	0.47	0.12	27.31
SSP	16.24	59.41	32.82	09.19	28.01	21.18	44.03	31.69	5.76	18.19
Carbonate hazard indices										
RSC	− 10.77	0.55	− 2.60	2.01	− 77.48	− 5.43	0.63	− 2.30	1.24	− 54.05
PI	41.87	77.33	57.50	7.07	12.30	47.21	74.27	57.95	5.75	9.93
MHR	39.07	472.00	52.29	54.84	104.87	38.21	53.31	45.60	2.67	5.86
Irrigation water quality index										
IWQI	48.00	89.00	76.16	10.59	13.91	59.00	92.00	80.91	6.69	8.26

Table 5. Sstatistical summary of irrigation water quality indices. Min, Max, SD, and CV denote minimum, maximum, standard deviation, and coefficient of variance (%), respectively

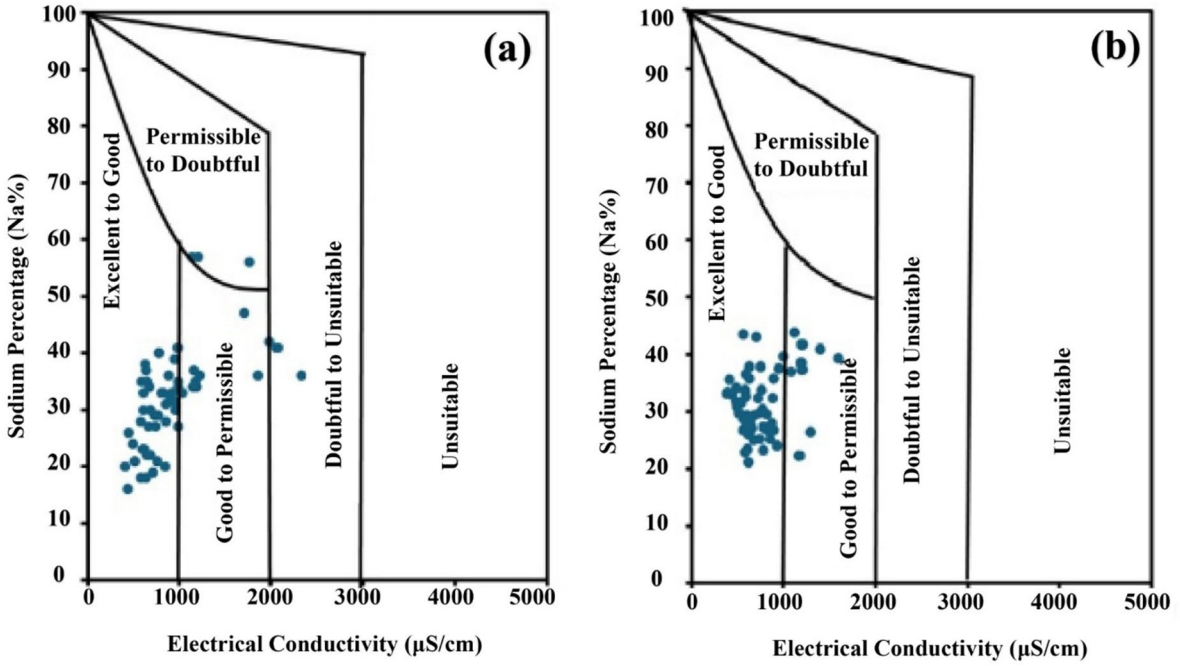


Fig. 4. Wilcox classification of groundwater based on sodium percentage ($\%Na^+$) for (**a**) pre-monsoon and (**b**) post-monsoon seasons.

*Sodium hazard indices (**$\%Na^+$**, **SAR, KR**, and **SSP**)*

Sodium hazard indices are essential for evaluating irrigation water suitability because excessive Na^+ in irrigation water can displace Ca^{2+} and Mg^{2+} from soil exchange sites, leading to clay dispersion, reduced permeability, poor aeration, and restricted root development. In semi-arid hard-rock aquifers, Na^+ enrichment is commonly associated with silicate weathering, cation exchange, evapoconcentration, and irrigation return flow. Therefore, Na^+-based indices are particularly relevant for assessing long-term soil–water interactions.

The $\%Na^+$, calculated using Eq. (2), ranges from 16.17–57.98% during pre-monsoon season and 21.10–43.82% during post-monsoon season (Table 5). According to the Wilcox classification (Table 4; Fig. 4), all samples in both seasons fall within the suitable category ($\%Na^+ < 60$), indicating low sodicity risk for irrigated soils. The higher upper range observed during pre-monsoon reflects evapoconcentration and enhanced ion exchange under declining groundwater levels, whereas the narrower post-monsoon range indicates partial dilution following recharge.

SAR values, computed using Eq. (3), vary from 0.61–5.08 in pre-monsoon and 0.86–3.16 in post-monsoon season (Table 5). Based on Richards' classification, all samples fall within the excellent category (*SAR* < 10),

suggesting negligible Na^+-induced soil dispersion risk in both seasons (Table 4). The persistently low *SAR* values, even where EC is moderate to high, indicate that Ca^{2+} and Mg^{2+} concentrations remain sufficient to counterbalance Na^+ effects and maintain soil permeability.

The combined influence of salinity and sodicity was evaluated using the USSL diagram[44] (Fig. 3). During pre-monsoon season, 51% of samples plot in the C2–S1 field and 47% in C3–S1, indicating medium to high salinity with low sodium hazard. Only a minor fraction (~ 4%) occurs in C3–S2 and C4–S1 fields. It reflecting localized salinity stress rather than sodicity constraints. In the post-monsoon season, samples are restricted to C2–S1 (59%) and C3–S1 (41%), confirming seasonal dilution of salinity without any significant shift in sodium hazard status. The consistent dominance of the S1 class across seasons demonstrates that salinity, rather than sodicity, is the primary irrigation limitation in the study area.

Kelly's ratio (*KR*), calculated using Eq. (4), ranges from 0.19–1.46 in pre-monsoon and 0.27–0.79 in post-monsoon season (Table 5). Based on Kelly's criterion (Table 4), 95% of pre-monsoon samples and 100% of post-monsoon samples are suitable for irrigation ($KR < 1$). The few pre-monsoon samples exceeding unity indicate localized Na^+ enrichment under evaporative conditions, which is effectively mitigated after monsoonal recharge.

SSP values, calculated using Eq. (5), range from 16.24–59.41 during pre-monsoon and 21.18–44.03 during post-monsoon season (Table 5). Based on SSP classification (Table 4), 77% of pre-monsoon samples are suitable, while 23% fall under unsuitable categories. In contrast, 85% of post-monsoon samples are classified as unsuitable despite lower absolute Na^+ concentrations. This apparent inconsistency reflects the greater sensitivity of *SSP* to relative ionic proportions rather than absolute Na^+ levels, suggesting that post-monsoon dilution of Ca^{2+} and Mg^{2+} can increase the relative dominance of Na^+ in solution.

Carbonate hazard indices (RSC, PI, and MHR)

Carbonate species (CO_3^{2-} and HCO_3^-) significantly influence irrigation water suitability by promoting the precipitation of Ca^{2+} and Mg^{2+} as carbonates, which in turn increases the relative dominance of Na^+ on soil exchange sites and may impair soil structure[49].

In the study area, *RSC* values range from − 10.47 to 0.55 meq/L during pre-monsoon season and from − 5.43 to 0.63 meq/L during post-monsoon season (Table 5). According to standard classification criteria, all samples (100%) in both seasons fall within the good category ($RSC < 1.25$ meq/L), indicating the absence of carbonate-induced sodicity risk (Table 4). The dominance of negative *RSC* values further suggests an excess of Ca^{2+} and Mg^{2+}, characteristic of silicate weathering–controlled hard-rock aquifers[50].

The *PI*, calculated using Eq. (7) following Doneen, integrates these parameters to evaluate potential long-term effects on soil hydraulic properties. *PI* values range from 41.87–77.33% during pre-monsoon season and from 47.21–74.27% during post-monsoon season (Table 5). Based on Doneen's classification (Table 4), 97% of pre-monsoon samples and 100% of post-monsoon samples fall within Class II (25–75%), indicating moderate permeability conditions. The Doneen diagram (Fig. 5) confirms that groundwater from both seasons predominantly occupies the Class II field, suggesting moderate but persistent permeability constraints rather than severe structural degradation.

The *MHR*, calculated using Eq. (8), assesses the relative dominance of Mg^{2+} over Ca^{2+} in irrigation water. Excess Mg^{2+} can adversely affect soil aggregation, reduce permeability, and suppress crop productivity[51]. Pre-

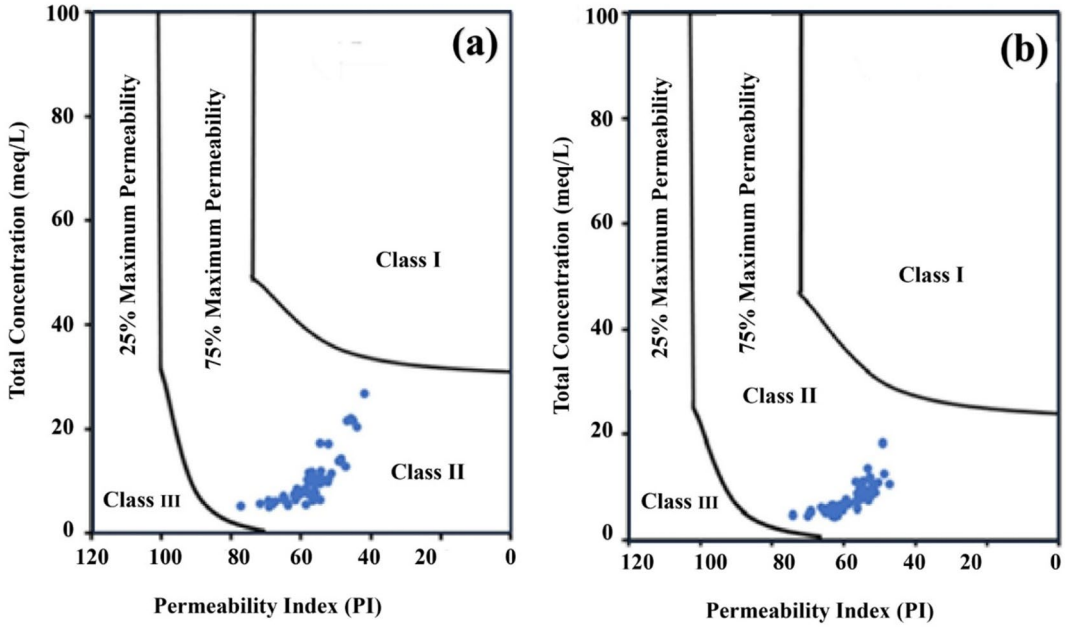


Fig. 5. Doneen's permeability index classification of groundwater for irrigation suitability during (**a**) pre-monsoon and (**b**) post-monsoon seasons.

monsoon *MHR* values range from 39.07 to 472, with 85% of samples classified as suitable (*MHR* < 50) and 15% as unsuitable (Tables 4 and 5). During post-monsoon season, *MHR* values range from 38.21 to 53.31, with 95% of samples falling within the suitable category. The improved post-monsoon suitability reflects dilution of Mg^{2+} concentrations and increased Ca^{2+} availability following recharge.

Irrigation water quality index (*IWQI*)

Seasonal variation of IWQI

To evaluate the integrated influence of key hydrochemical parameters controlling irrigation suitability (*EC*, Na^+, HCO_3^-, Cl^-, and *SAR*), the *IWQI* was computed using Eqs. 9–11.

During the pre-monsoon season, *IWQI* values range from 48 to 89, whereas post-monsoon values range from 59 to 92 (Table 5). Based on established *IWQI* classification criteria[31], groundwater quality is categorized as excellent (*IWQI* > 85), good (70–85), poor (55–70), very poor (40–55), and unsuitable (< 40).

Accordingly, during pre-monsoon season, 23% of samples fall within the excellent category, 62% are classified as good, 55% as poor, and 10% as very poor (Table 4). In contrast, post-monsoon results indicate a clear improvement, with 39% of samples classified as excellent, 57% as good, and 4% as poor. Notably, no samples fall within the very poor or unsuitable categories during post-monsoon period.

Spatial distribution of IWQI

The spatial distribution of *IWQI* values across the study area is illustrated in Fig. 6, highlighting zones of varying irrigation suitability. During pre-monsoon season (Fig. 6a), approximately 23.73% of the area is classified as good, 65.01% as moderate, 8.41% as poor, and 2.85% as very poor. Areas with lower *IWQI* values are primarily concentrated in the central and eastern parts of the study area, where intensive groundwater abstraction, limited recharge, and evapoconcentration effects are dominant.

In the post-monsoon season (Fig. 6b), a marked spatial improvement is observed, with 34.69% of the area classified as good, 51.36% as moderate, and 13.95% as poor, and 7.70% as very poor. The complete disappearance of the very poor category indicates effective dilution and partial flushing of accumulated salts following monsoonal rainfall. However, the persistence of poor-quality zones in localized areas suggests that lithological controls, aquifer thickness, and land-use practices continue to constrain full geochemical recovery.

Multivariate statistical interpretation

Correlation matrix interpretation

Pearson's correlation analysis was conducted for pre- and post-monsoon datasets to examine interrelationships among irrigation water quality indices and to identify dominant controls on groundwater suitability for irrigation (Table 6).

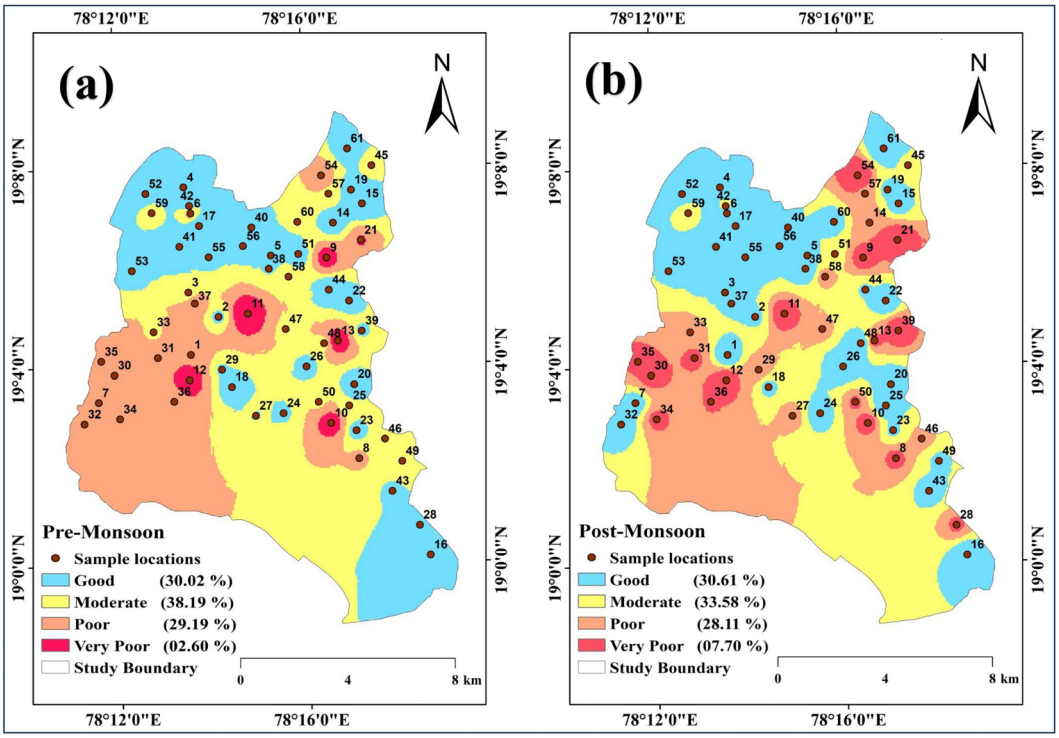


Fig. 6. Spatial distribution of irrigation water quality index (*IWQI*) for (**a**) pre-monsoon and (**b**) post-monsoon seasons.

	EC	%Na⁺	SAR	KR	MHR	PI	SSP	RSC	IWQI
(a)									
EC	1.00								
%Na⁺	0.60**	1.00							
SAR	0.83**	0.92**	1.00						
KR	0.55**	0.97**	0.92**	1.00					
MHR	-0.04	0.06	-0.01	0.03	1.00				
PI	-0.75**	-0.37**	-0.57**	-0.30**	0.16	1.00			
SSP	0.60**	1.00**	0.92**	0.97**	0.05	-0.36**	1.00		
RSC	-0.76**	-0.09	-0.36**	0.01	0.11	0.83**	-0.09**	1.00	
IWQI	-0.83**	-0.38**	-0.67**	-0.43**	0.08	0.53**	-0.38**	0.61**	1.00
(b)									
EC	1.00								
%Na⁺	0.26*	1.00							
SAR	0.74**	0.82**	1.00						
KR	0.29*	1.00**	0.82**	1.00					
MHR	0.15	0.01	0.09	0.00	1.00				
PI	-0.67**	0.17	-0.31**	0.17	-0.39**	1.00			
SSP	0.25*	1.00**	0.81**	0.99**	0.00	0.18	1.00		
RSC	-0.64**	0.39**	-0.13	0.38**	-0.19	0.90	0.40**	1.00**	
IWQI	-0.39**	-0.06	-0.30**	-0.07	-0.10	0.04	-0.05**	0.07	1.00**

Table 6. Pearson's correlation coefficients matrix among irrigation water quality indices for (a) pre-monsoon and (b) post-monsoon seasons. *Correlation is significant at the 0.05 level; **Correlation is significant at the 0.01 level.

Pre-monsoon: During pre-monsoon season, strong to very strong positive correlations among Na^+-related indices indicate a coherent sodicity control on irrigation water quality (Table 6). EC shows strong positive correlations with $\%Na^+$ (r = 0.60), SAR (r = 0.83), and SSP (r = 0.60). This suggests that increased salinity is closely associated with enhanced sodium hazard under evapoconcentrated conditions typical of semi-arid hard-rock aquifers[36]. *%Na*$^+$ exhibits very strong correlations with *SAR* (*r* = 0.92), *KR* (*r* = 0.97), and *SSP* (*r* = 1.00), confirming the central role of Na^+ enrichment in governing sodicity-related indices. Similarly, strong correlations among *SAR, KR*, and *SSP* reflect their shared dependence on Na^+ dominance relative to Ca^{2+} and Mg^{2+}.

Permeability-related indices also show meaningful associations. *PI* correlates positively with *RSC* (*r* = 0.83), while *RSC* shows a moderate positive correlation with *IWQI* (*r* = 0.61), indicating that carbonate balance and permeability conditions influence integrated irrigation suitability. In contrast, EC exhibits strong negative correlations with *PI* (r = –0.75), *RSC* (*r* = –0.76), and *IWQI* (r = –0.83), demonstrating the adverse influence of salinity on soil permeability and overall irrigation quality. *SAR* also shows a significant negative correlation with *IWQI* (*r* = –0.67), reinforcing the detrimental impact of sodium hazard on irrigation suitability.

Post-monsoon: After monsoonal recharge, correlation strengths generally decrease due to dilution, although key interrelationships persist (Table 6). EC remains strongly correlated with *SAR* (*r* = 0.74), indicating continued salinity control on sodium hazard. Na^+-related indices retain strong interdependence, with *%Na*$^+$ showing near-perfect correlations with *KR* (*r* = 1.00) and *SSP* (*r* = 1.00), and a strong correlation with *SAR* (*r* = 0.82). These relationships confirm the persistence of Na^+ dominance following recharge.

Among permeability-related indices, *PI* shows a very strong positive correlation with *RSC* (*r* = 0.90), suggesting continued carbonate control on permeability conditions. Negative correlations of EC with *PI* (*r* = –0.67) and *RSC* (*r* = –0.64) persist, indicating that salinity continues to suppress permeability despite seasonal dilution.

PCA and dominant hydrochemical control group

Principal component analysis (PCA) was applied to identify the dominant hydrogeochemical processes controlling irrigation suitability during pre- and post-monsoon seasons (Table 7).

Pre-monsoon: Three principal components (PCs) with eigenvalues > 1 explain 92.21% of the total variance (cumulative eigenvalue = 8.30), indicating that irrigation water quality during dry-season conditions is governed by a limited number of dominant processes.

PC1 accounts for 44.89% of the variance (eigenvalue = 4.04) and shows strong positive loadings for *%Na*$^+$ (0.98), *SAR* (0.88), *KR* (0.99), and *SSP* (0.98), identifying sodium hazard as the primary controlling factor. This component reflects evapoconcentration-driven Na^+ enrichment under prolonged residence time and limited recharge conditions, adversely affecting soil permeability and irrigation suitability.

PC2 explains 36.12% of the variance (eigenvalue = 3.25) and is characterized by strong positive loadings of *PI* (0.86) and *RSC* (0.97) and a strong negative loading of EC (–0.84). This component represents a permeability–

Parameters	Pre-monsoon			Post-monsoon		
	PC1	PC2	PC3	PC1	PC2	PC3
EC	0.48	**− 0.84**	0.02	0.35	**− 0.73**	− 0.49
%Na^+	**0.98**	− 0.17	0.04	**0.99**	0.10	0.01
SAR	**0.88**	− 0.47	− 0.01	**0.86**	− 0.37	− 0.31
KR	**0.99**	− 0.11	− 0.01	**0.99**	0.10	− 0.02
MHR	0.04	0.08	**0.99**	0.06	− 0.49	0.23
PI	− 0.21	**0.86**	0.12	0.08	**0.97**	0.05
SSP	**0.98**	− 0.16	0.03	**0.99**	0.11	0.01
RSC	0.10	**0.97**	0.03	0.31	**0.89**	0.17
IWQI	− 0.33	**− 0.76**	0.01	− 0.05	0.01	**0.92**
Eigenvalue	4.04	3.25	1.01	3.94	2.67	1.27
% of Variance	44.89	36.12	11.20	43.73	29.69	14.15
Cumulative %	44.89	81.02	92.21	43.73	73.418	87.57

Table 7. Principal component analysis (PCA) loadings for irrigation water quality indices. Bold denotes PCA > 0.70.

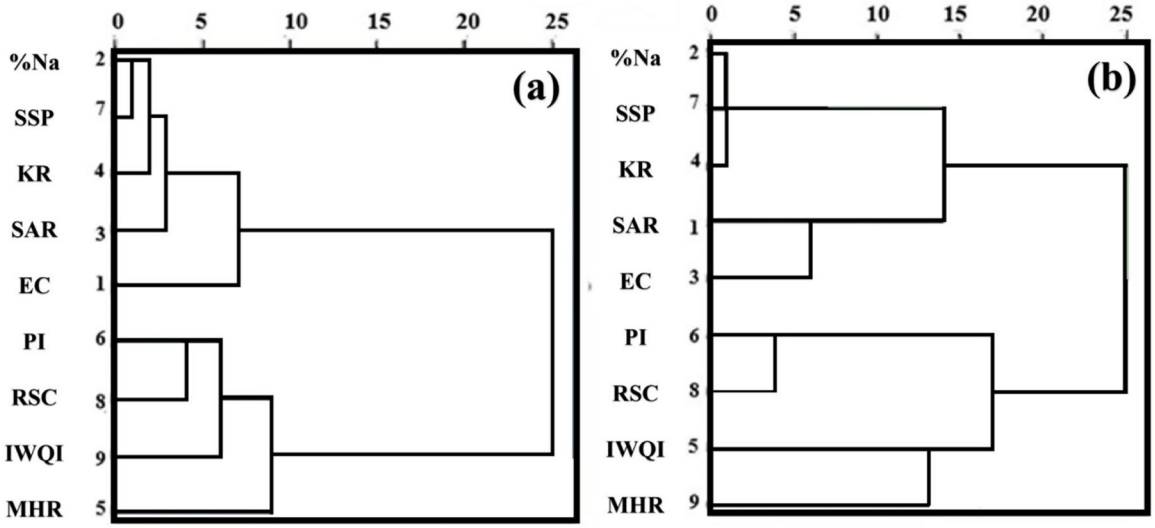


Fig. 7. Hierarchical cluster dendrogram of groundwater samples based on irrigation water quality indices.

salinity contrast, where lower salinity is associated with improved soil permeability and reduced carbonate hazard, consistent with correlation results.

PC3 accounts for 11.20% of the variance (eigenvalue = 1.01) and is dominated by *MHR* (0.99), indicating that magnesium hazard acts as an independent secondary control on irrigation suitability, particularly in areas with elevated Mg^{2+} relative to Ca^{2+}.

Post-monsoon: In post-monsoon season, three PCs explain 87.57% of the total variance (cumulative eigenvalue = 7.88), reflecting dilution-induced variability while retaining the same fundamental controls.

PC1 (43.73% variance; eigenvalue = 3.94) remains dominated by *%Na*$^+$ (0.99), *SAR* (0.86), *KR* (0.99), and *SSP* (0.99), confirming the persistence of sodium hazard as the primary determinant of irrigation water quality after recharge. PC2 (29.69% variance; eigenvalue = 2.67) shows high positive loadings for PI (0.97) and RSC (0.89) and a negative loading for EC (-0.73). This indicates partial recovery of permeability and carbonate balance under diluted conditions. PC3 (14.15% variance; eigenvalue = 1.27) is characterized by a strong positive loading of *IWQI* (0.92), suggesting improved coherence between individual indices and integrated irrigation suitability following recharge.

HCA grouping and irrigation suitability

Hierarchical cluster analysis (HCA) was performed to identify natural groupings among irrigation water quality indices and their collective influence on irrigation suitability (Fig. 7).

Pre-monsoon: Two major clusters are identified (Fig. 7a). Cluster 1 comprises *%Na*$^+$, *SSP*, *KR*, *SAR*, and EC, forming at a lower rescaled distance (~ 7.5), reflecting strong coupling between sodium hazard and salinity

under evapoconcentrated conditions. Cluster 2 includes *PI*, *RSC*, *IWQI*, and *MHR*, merging at a slightly higher rescaled distance (~ 9.5), indicating secondary control by permeability, carbonate balance, and magnesium hazard. The relatively low linkage distances reflect strong hydrochemical coherence during the dry season.
Post-monsoon: In the post-monsoon period (Fig. 7b), the same two clusters persist, but with increased rescaled distances. Cluster 1 (*%Na*$^+$, *SSP*, *KR*, *SAR*, and EC) forms at ~ 14, indicating weakened sodium–salinity coupling due to dilution. Cluster 2 (*PI, RSC, IWQI*, and *MHR*) merges at ~ 16, suggesting greater dispersion among permeability-related indices after recharge. These changes reflect reduced chemical coherence without altering the underlying hydrogeochemical structure.

Multiple linear regression (MLR) and predictive controls on IWQI
Seasonal multiple linear regression (MLR) analysis was applied to quantify the predictive influence of key hydrochemical parameters (*EC*, *Na*$^+$, *HCO*$_3^-$, *Cl*$^-$, and *SAR*) on *IWQI* and to validate the dominant controls identified through correlation analysis, PCA, and HCA.
Pre-monsoon: During the pre-monsoon season, the MLR model exhibits strong predictive capability (Table 8), with a multiple correlation coefficient (*R*) of 0.85, R^2 of 0.71, and adjusted R^2 of 0.69, indicating that approximately 71% of the variability in *IWQI* is explained by the selected predictors. This high explanatory power is consistent with the strong interdependence among Na^+-related indices observed in the correlation matrix and the dominance of sodium hazard identified in PCA (PC1) and HCA (Cluster 1).

Model robustness is further supported by a strong observed–predicted correlation (r = 0.84; 95% CI^- = 0.75–0.90) and a statistically significant regression (F statistic = 27.42; $p < 0.01$) (Fig. 8a). Low prediction errors (*RMSE* = 0.34; *MAE* ≈ 0.00) indicate reliable model performance under dry-season conditions.

Standardized regression coefficients identify *SAR* (β = 0.84) and Na^+ (β = –0.70) as the dominant predictors of *IWQI*, followed by EC (β = 0.16), while HCO_3^- (β = 0.01) and Cl^- (β = –0.06) make minor contributions (Table 8). These results corroborate multivariate findings that highlight Na^+-related processes as the principal control on irrigation water quality during the pre-monsoon period. The regression relationship is expressed as:

$$IWQI = 83.04 + 0.16EC - 0.70Na^+ + 0.01HCO_3^- - 0.06Cl^- + 0.84SAR \quad (16)$$

Hydrochemically, pre-monsoon conditions are characterized by lower groundwater levels, enhanced evaporation, and limited recharge, resulting in higher ionic concentrations and stronger ion–ion interactions. These conditions promote a more coherent hydrochemical framework and yield a predictable IWQI response dominated by sodium and salinity stress.
Post-monsoon: In contrast, the post-monsoon MLR model shows reduced predictive strength due to increased hydrochemical heterogeneity following recharge. The multiple correlation coefficient decreases to R = 0.57, with R^2 = 0.33 and adjusted R^2 of 0.27, indicating that only 33% of *IWQI* variability is explained by the selected parameters (Table 8). The observed–predicted correlation is moderate (r = 0.47; 95% CI^- = 0.25–0.65), reflecting increased uncertainty (Fig. 8b). Although the regression remains statistically significant (F statistic = 6.34; $p < 0.01$), predictive accuracy declines, as indicated by higher *RMSE* (0.57) and *MAE* (0.97) values.

Standardized coefficients reveal a seasonal shift in dominant controls. EC emerges as the strongest positive predictor (β = 1.20), while Na^+ (β = –0.87), Cl^- (β = –0.78), and HCO_3^- (β = –0.60) exert stronger negative influences. *SAR* (β = 0.60) shows a reduced and less consistent role compared to pre-monsoon conditions. The post-monsoon regression equation is given as:

From MLR	Pre-monsoon	Post-monsoon
R	0.85	0.57
R^2	0.71	0.33
Adjusted *R*	0.69	0.27
F-statistic	27.42	6.34
P-value	0.00	0.00
SEE	5.92	5.73
Intercept	83.04	82.32
β EC	0.16	1.20
β Na^+	− 0.70	− 0.87
β HCO_3^-	0.01	− 0.60
β Cl^-	− 0.06	− 0.78
β *SAR*	0.84	0.60
β *RMSE*	0.34	0.57
β *MAE*	0	0.97
From Fig. 8		
r	0.84	0.47
95% confidence interval (Cl^-)	0.75–0.90	0.25–0.65

Table 8. Comparison between observed and MLR-predicted IWQI values and associated performance statistics for (a) pre-monsoon and (b) post-monsoon seasons.

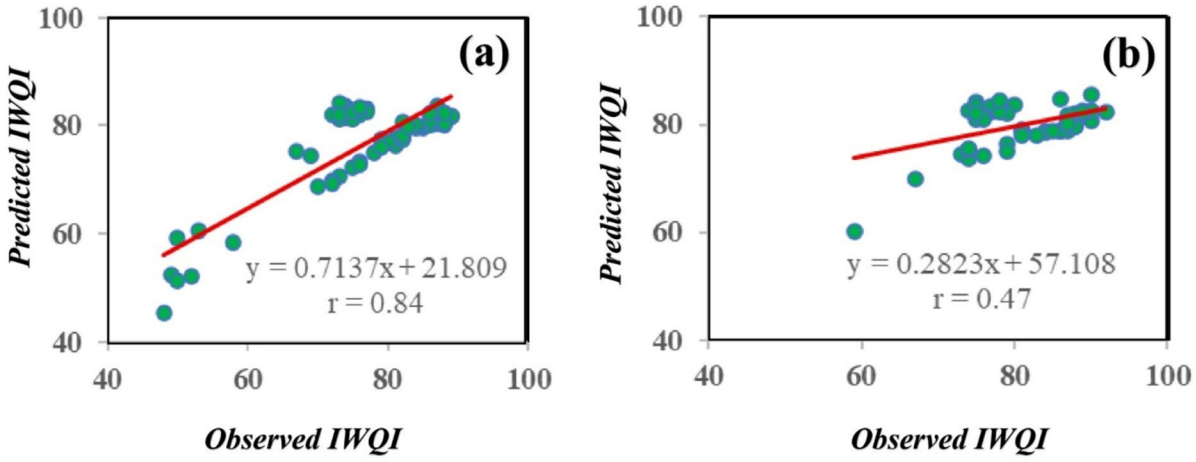


Fig. 8. Relationship between observed and MLR-predicted Irrigation Water Quality Index (*IWQI*) values for (**a**) pre-monsoon and (**b**) post-monsoon seasons.

$$IWQI = 82.32 + 1.20EC - 0.87Na^{+} - 0.60HCO_{3}^{-} - 0.78Cl^{-} + 0.60SAR \quad (17)$$

These changes reflect the effects of monsoonal recharge, which dilutes groundwater chemistry and disrupts established ionic relationships. Monsoonal recharge also introduces additional variability through surface runoff, irrigation return flow, and shallow aquifer mixing. Under these conditions, EC acts as a more sensitive indicator of residual salinity variability, while sodium- and carbonate-related controls become less coherent-consistent with reduced PCA loadings and increased HCA linkage distances.

Applications, limitations, and management considerations

Applications

This study proposes an integrated framework for evaluating irrigation water suitability in semi-arid regions by combining irrigation indices with multivariate statistical tools. The approach identifies key hydrochemical controls affecting irrigation sustainability, particularly sodium hazards influencing soil permeability. The results support crop selection, irrigation planning, and soil–water management. The framework is transferable to similar hydrogeological regions and contributes to Sustainable Development Goal 3 and Sustainable Development Goal 6 by promoting safe water use and sustainable water resource management.

Limitations

The present analysis focuses on major irrigation water quality indices based on two seasonal datasets (pre- and post-monsoon), which capture broad seasonal contrasts but do not fully represent short-term hydrological variations that may influence salinity and sodicity dynamics. Groundwater table fluctuation data were not available due to the absence of dedicated monitoring wells and consistent static water level records in the study area; therefore, the relationship between groundwater level fluctuations and irrigation water quality could not be evaluated in this study. In addition, localized aquifer heterogeneity and site-specific anthropogenic influences may not be fully represented at all locations. Consequently, application of this framework in other regions may require consideration of local lithology, recharge characteristics, groundwater level conditions, and agricultural practices.

Management considerations

Groundwater classified as excellent to good can be used for irrigation without restriction, whereas marginal-quality water may require blending, soil amendments, or the cultivation of salt-tolerant crops. Areas affected by elevated sodium hazard should adopt management practices such as gypsum application, improved drainage, and regulated irrigation to prevent soil structural degradation. Given the dominance of sodicity over salinity identified in this study, long-term management should prioritize maintaining a favourable Ca^{2+}–Na^{+} balance and preserving soil permeability. Seasonal variability, particularly post-monsoon dilution effects, underscores the need for adaptive, site-specific irrigation management strategies.

Conclusions

This study assessed groundwater suitability for irrigation in a semi-arid region of Telangana, India, using multiple irrigation water quality indices integrated with multivariate statistical and regression analyses. The results indicate that groundwater is generally suitable for irrigation, with 85% of pre-monsoon samples and 96% of post-monsoon samples classified as excellent to good based on the *IWQI*. The post-monsoon improvement reflects the dilution effect of monsoonal recharge on dissolved salts and sodium concentrations:

a) Pre-monsoon groundwater is characterized by relatively higher salinity and sodium hazards, as evidenced by elevated *EC*, *SAR*, *%Na*$^+$, *KR*, and *SSP* values. In contrast, post-monsoon recharge improves soil permeability (*PI*) and overall irrigation suitability. Pearson's correlation analysis reveals strong interdependence among Na^+-related indices and significant inverse relationships between EC, *PI*, and *IWQI*, confirming the dominant role of Na^+–EC interactions in controlling irrigation water quality,
b) PCA explains 87–92% of the total variance across both seasons and identifies sodium hazard, salinity, carbonate balance, and magnesium effects as the principal governing processes. HCA corroborates these findings by consistently separating Na^+–EC–related indices from permeability- and carbonate-controlled parameters, indicating a hierarchical control structure on irrigation suitability,
c) Multivariate statistical analysis indicated that sodium hazard and salinity collectively govern irrigation suitability. The results further revealed a clear seasonal shift in predictive stability, with stronger and more reliable prediction during the pre-monsoon season ($R^2 = 0.71$) and reduced predictability during the post-monsoon season ($R^2 = 0.33$) due to dilution and hydrochemical mixing following monsoonal recharge. These findings indicate that long-term irrigation sustainability in the study area primarily depends on managing sodicity through appropriate soil and water management practices, and
d) The findings demonstrate that sodium accumulation, rather than salinity alone, constitutes the primary constraint on irrigation suitability in the study area. The pronounced seasonal contrasts underscore the importance of season-specific monitoring and adaptive management strategies to ensure sustainable groundwater use and long-term soil productivity in semi-arid environments.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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